



Customer Feedback Intelligence

Contract kickoff — scope, success criteria, and the boundary of honesty defined before any numbers exist.

APPLIED AI PROJECT

E-COMMERCE REVIEWS

The Situation



An online women's fashion retailer has **tens of thousands of free-text reviews** and no structured way to use them.



The challenge: turn raw text into action — without setting up a labeling operation first.

Support Team

Responds to each review from scratch, with no context on themes or similar cases.

Merchandising Team

No quick read on what customers are raising about product and catalog.

The Three Personas



Customer Support Lead

Priority: Response quality and speed.
Wants to triage the topic, read the sentiment, and see similar past cases to respond quickly and consistently.



Merchandising / Product

Priority: Product and catalog decisions.
Needs an honest read on what customers are saying: quality, fit, style, comfort, value, and appearance.



Senior AI / ML Engineer

Priority: Embedding pipeline, evaluation rigor, and the honesty frontier. Defensible metrics, an honest baseline, a shared vector set between evaluation and product.

The Four Deliverables

A small, reproducible system that delivers the brief's capabilities end to end — accessed through a functional **Review Reply Assistant** and documented in two notebooks.



Embeddings

Every review encoded with a single shared encoder.



2D Visualization

Projection of embeddings into two dimensions for visual interpretation.



Categorization

Theme assignment: quality, fit, style, comfort, value, or appearance.



Similarity Search

Given a review, returns the nearest matches for consistent responses.

The Boundary of Honesty

This is the line the project keeps — because it is the easiest thing to distort.

→ No ground truth for topics

The dataset has no topic labels. Topic assignment is an **unsupervised triage aid**, never a measured classifier. We never report "topic accuracy."

→ What we actually measure

Whether the *embedding space* is meaningful, using the labels the dataset truly has: **Recommended IND** (binary sentiment) and **Department Name** (product area).

→ Embeddings are not assumed to be the winner

For short binary text, a bag-of-words model is genuinely strong. If it beats embeddings, we say so.



The Acceptance Bar

Defined before any number. The system is "good enough to build on" if **both** conditions are met:

① Semantic Structure

Under a linear probe, the embeddings **at least match** the TF-IDF baseline on Department Name (a semantic label), within a small margin.

② Useful Retrieval

Nearest-neighbor retrieves reviews from the same department **above random chance** — target: $\geq 2\times$ the chance rate at $k=3$.

- 📄 Deliberately outside the bar: beating TF-IDF on sentiment. Bag-of-words is a strong baseline on short text — and, as the results show, it wins there.

The Dataset



Women's E-Commerce Clothing Reviews – Kaggle

Public dataset of **23,486** real, anonymized reviews, with 10 columns.

Source: [nicapotato](#).

Product input

Only **Review Text** — no other column is used as a feature by the system.

Evaluation labels

Recommended IND (binary sentiment) and **Department Name** (product area) — used exclusively for evaluation, never as input.